

Omar Ibrahim

Jr Penetration Tester & Ethical Hacker

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SUMMARY

Aspiring Junior Penetration Tester and IT student with hands-on offensive security experience through TryHackMe, Hack The Box, and PortSwigger Web Security Academy. Focused on web application penetration testing with practical skills in vulnerability assessment, exploitation, and web enumeration. Passionate about offensive security and committed to developing expertise in ethical hacking and penetration testing methodologies

EDUCATION

Bachelor of Technology – Information Technology

New Cairo Technological University

Expected 2029

Cairo, Egypt

High School Diploma – Embedded Systems & AI

Misr International Computer & AI Academy (MICA)

Jun 2025

Giza, Egypt

- Ranked 3rd Nationally across the Arab Republic of Egypt

EXPERIENCE

Embedded Software Developer | LABstar

Aug 2023 – Jun 2025

Cairo

Egypt

- Enforced Role-Based Access Control (RBAC) across multi-user industrial applications, ensuring data integrity and least-privilege security principles
- Executed RFID Authentication for time-limited floor access control on elevator systems, overseeing secure hardware-to-board communication and user authorization
- Created a cross-platform IoT controller (Flutter + ESP32 via Wi-Fi) with real-time hardware synchronization and low-latency command execution — skills directly applicable to IoT security assessments
- Developed inventory and sales management systems with SQL database backends and automated audit trails

Python Developer Intern | CodeAlpha

Oct 2025 – Nov 2025

Remote

Indian

- Enhanced Python automation scripts utilizing OS, Shutil, and Regex modules, strengthening security tooling and workflow automation capabilities
- Expanded logic-driven applications leveraging advanced data structures and robust error-handling best practices

Penetration Testing & Application Security Trainee | NTI

Aug 2026 – Sep 2026

Online

Cairo

- Trained in vulnerability assessment, WordPress hardening, penetration testing methodologies, cloud security auditing, and application security
- Gained practical exposure to log analysis, threat hunting, and security awareness campaign planning
- Completed a real-world capstone project and freelance security engagement
- Built a professional freelance portfolio and personal brand for cybersecurity service delivery (Upwork, HackerOne)

PROJECTS

Penetration Testing Labs — TryHackMe – Top 8% Ranking

- Completed rooms covering network enumeration, privilege escalation, web application attacks, and CTF challenges
- Applied **Nmap**, **Metasploit**, **Gobuster**, and Linux exploitation techniques in simulated adversarial environments
- **Ranked 4th at HITU IEEE CTF** — competed solo against teams of 3–5 members, demonstrating cross domain self-sufficiency under competitive pressure

IBM Ethical Hacking Capstone — IBM Cybersecurity Analyst Professional Certificate

- Conducted vulnerability scans using **OpenVAS** and analyzed CVE findings to prioritize remediation strategies
- Executed exploitation exercises with Metasploit Framework in controlled lab environments

- Completed 14-course program covering the full security lifecycle: threat intelligence, SIEM, incident response, and network defense

Smart Elevator Interface System — LABstar – Embedded Security

- Applied **RFID-based** authentication with time-limited floor access, overseeing hardware-to-board secure communication and user authorization logic

Electronics Management System — LABstar – Application Security

- Enforced strict RBAC to secure multi-user collaboration across a **SQL-backed** inventory system, preventing unauthorized data access and maintaining audit trails

Bank Management System | Python, Tkinter, CustomTkinter

- Created multi-user GUI with RBAC for Customers, Tellers, and Managers; applied encrypted-style permission levels and transaction logging for accountability

Real-Time Sensor Data Pipeline | Python, numpy, matplotlib, sqlite

- Developed data ingestion pipeline with serial communication, **SQLite** storage, and **Matplotlib** visualization — applicable to security monitoring and anomaly detection

Cinema Management System | C

- File-based system for movie management, ticket booking, hall and seat reservation
- Applied modular programming, file handling, and team collaboration

Morse Code Converter | x86-64 Assembly

- Terminal-based converter using Win32 API only, no external libraries
- Built O(1) lookup table, manual memory management, and direct console I/O
- Demonstrates low-level binary thinking essential for reverse engineering

Smart Home with Voice Control (*Graduation Project*) | Flutter, Firebase, PCB

- Voice-controlled smart home managed via **Flutter** mobile app connected to Firebase
- Led the team as Team Leader — task distribution, **PCB design**, and documentation

KeyGarden – Password Generator | Python, CustomTkinter

- GUI-based password generator with auto, text-based, passphrase, and batch export modes
- Built for security-conscious users needing strong, flexible password generation

Universal Number System Converter | Python, Tkinter

- Calculator and converter across Binary, Hex, Decimal, Octal, and ASCII
- Useful for low-level programming and security work involving data representation

SKILLS

Languages: C, Python, JS, SQL, Bash, x86-64 Assembly, Powershell

Operating Systems: Linux, Windows

Security Testing: Owasp Top 10, Manual Testing, Reconnaissance, Enumeration, Exploitation, Networking Fundamentals

Tools: Metasploit, OpenVAS, Gobuster, Nmap, Wireshark, Burp Suite, IDA, SQLmap, Nessus

Other: MS Office, Git/GitHub, Firebase, ESP32/ESP8266, Flutter, Oracle SQL, MySQL

LEADERSHIP

Technical Project Lead – Graduation Project

2025

MICA

- Directed a multidisciplinary team to develop a Voice-Controlled Smart Home integrating Flutter, Firebase, and ESP8266
- Managed task timelines, technical documentation, and cross-functional coordination across engineering workstreams

Technical Coordinator, RoboCup

2025

MICA

- Led the Line Follower category, overseeing technical inspections for competing robotics teams
- Mentored participants on sensor calibration and logic optimization techniques

CERTIFICATES

Cybersecurity Analyst Professional Certificate (14 courses) — IBM

Jan 2026

Ethical Hacking with OpenVAS & Metasploit — IBM

Jan 2026

Embedded Systems using C — EDUCBA

Jan 2026

Database Programming with SQL — Oracle
Exploring C — University of Michigan
Boost Your Productivity with AI — Google
Introduction to AI / AI Concepts — Microsoft
C Programming — Programize

Dec 2024
Apr 2025
Aug 2024
Jun 2024
Feb 2023